A Convenient Procedure for the Preparation of Potassium Trioxalatoferrate (III)

An experiment involving the preparation and characterization of potassium trioxalatoferrate (III), $K_3Fe(C_2O_4)_3$, is included in a recent general chemistry laboratory manual. This compound is well suited for study in general chemistry since it has several desirable properties. It has an attractive green color; it is only sparingly soluble in water at 0° C, whereas it is quite soluble in hot water. Relatively large crystals can be grown without much difficulty. It is possible to quantitatively analyze it for water, iron, and oxalate by standard procedures which are frequently taught in beginning courses.

The literature methods for the preparation of this compound involve several steps which include time-consuming digestions and evaporations. The compound can be quickly and easily prepared in the following manner: Add 8.0 ml of a ferric chloride solution (0.40 g $FeCl_3$/ml—a solution which is commercially available) to a hot solution containing 12 g of $K_2C_2O_4 \cdot H_2O$ in 20 ml of water. Cool the solution to 0° C and keep it at this temperature until crystallization is complete. The product can be collected. However, it is convenient to recrystallize it at this point by decanting off the mother liquor and redissolving the product in 20 ml of warm water. The product can be precipitated by cooling at 0° C, collected by filtration, washed with a small volume of ice water, and air dried. This preparation was carried out by 250 students as their first general chemistry laboratory experiment and most students achieved yields of dry recrystallized product of better than 50% (more than 5 g).

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